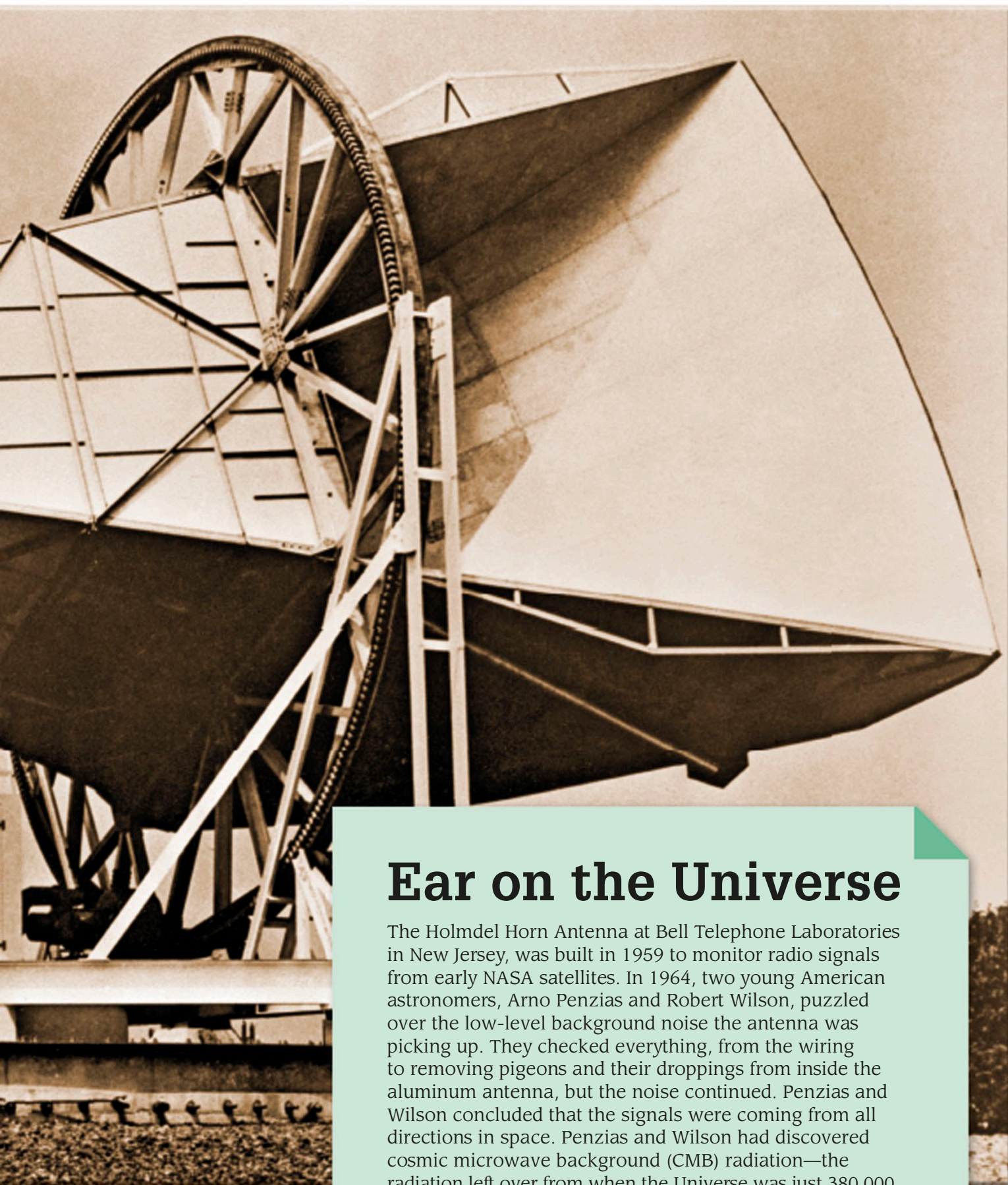




Ear on the Universe

The Holmdel Horn Antenna at Bell Telephone Laboratories in New Jersey, was built in 1959 to monitor radio signals from early NASA satellites. In 1964, two young American astronomers, Arno Penzias and Robert Wilson, puzzled over the low-level background noise the antenna was picking up. They checked everything, from the wiring to removing pigeons and their droppings from inside the aluminum antenna, but the noise continued. Penzias and Wilson concluded that the signals were coming from all directions in space. Penzias and Wilson had discovered cosmic microwave background (CMB) radiation—the radiation left over from when the Universe was just 380,000 years old. This was compelling evidence supporting the Big Bang theory of how the Universe began.